September 23, 2026

Prof. George Arhonditsis, Ph.D.
Editor-in-Chief, Ecological Informatics
Department of Physical and Environmental Sciences
University of Toronto
Toronto, Ontario, Canada

SUBJECT: Submission of Original Research Article to Ecological Informatics
Manuscript Title: "Integrating Multimodal Earth Observations and Hydrological Regularization for Spatiotemporal Wetland Water Mapping in the Indus Delta"

Dear Professor Arhonditsis,

Please consider the enclosed original research article entitled "Integrating Multimodal Earth Observations and Hydrological Regularization for Spatiotemporal Wetland Water Mapping in the Indus Delta" for publication in Ecological Informatics.

Coastal deltaic wetlands represent dynamic, ecologically fragile ecosystems that are increasingly threatened by compounding climatic anomalies and upstream freshwater reductions. In data-scarce deltas such as the Indus River Basin Delta—home to the world's largest arid-climate Avicennia marina mangrove forest—spatiotemporal ecological monitoring is severely hampered by two fundamental challenges: severe spatial resolution disparity between satellite observations (10 m) and macro-scale hydroclimatological drivers (4 km), and the inability of conventional machine learning models to respect physical conservation principles, often leading to unrealistic water-budget predictions and overfitting to static historical priors.

To address these challenges, we present the Multimodal Flood Physics-Regularized Transformer (MFPIT), a scientific machine learning framework designed specifically for environmental and ecological informatics. MFPIT introduces a directed Multi-Fidelity Cross-Attention (MF-CA) architecture that fuses multi-resolution Earth observation tokens with hydroclimatological fluxes under linear computational complexity. Furthermore, we incorporate an auxiliary surrogate mass-balance regularization term during backpropagation, reducing water-budget inconsistency by over 99% and anchoring neural predictions to physical conservation laws.

Evaluated over a comprehensive 23-year spatiotemporal archive (2001–2023) under a strict target-leakage-prevented benchmark (JRC=0), MFPIT establishes state-of-the-art performance across both hard boundary segmentation (Test IoU = 0.7102, F1 = 0.8306) and probabilistic dynamic ranking under severe class imbalance (PR-AUC = 0.9141, ROC-AUC = 0.9296), while demonstrating superior probability calibration (ECE = 0.0150, Brier score = 0.1057). Practically, this framework delivers an ecologically regularized decision-support methodology for estuarine conservation, mangrove habitat protection, environmental flow allocation below Kotri Barrage, and coastal flood-resilience planning.

Compliance with Ecological Informatics Author Guidelines:

- • Scope Alignment: Directly contributes to computational ecology, environmental informatics, big data Earth observation fusion, and ecosystem modeling.
- • Graphical Abstract: A mandatory standalone visual summary (4746 × 2196 pixels, landscape format) is provided.
- • Highlights: 5 concise highlight statements strictly conforming to the 85-character maximum per bullet are included.
- • Open Science & Reproducibility: Complete PyTorch source code, environment specifications, and cloud-native Google Earth Engine scripts are openly available at https://github.com/mirzamuzzamilbaig/MFPIT.
- • Mandatory Declarations: CRediT authorship contribution statement, Declaration of Competing Interest (no conflicts), and Generative AI disclosure are incorporated.

This manuscript represents original work, has not been published previously, and is not under consideration for publication elsewhere. The author has reviewed and approved the manuscript for submission.

Thank you for your time and consideration of our work.

Sincerely,

Mirza Muhammad Muzzamil
Department of Computer and Information Systems Engineering (CISE)
NED University of Engineering and Technology
University Road, Karachi 75270, Pakistan
Email: mirzamuzzamil@neduet.edu.pk